

CyberInvestigator

AI Fraud Network Intelligence Platform

“See the fraud network, not just the transaction.”

Theme — AI for Digital Public Safety:

Defeating Counterfeiting, Fraud & Digital Arrest Scams

Technical & Product Submission Document

Submitted for ET AI Hackathon 2.0

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1. Executive Summary

CyberInvestigator is an AI investigation platform built for law-enforcement agencies and financial-crime units. Rather than acting as another fraud classifier that scores individual transactions, it treats fraud as what it actually is in the real world — a coordinated network. Given raw transaction and complaint data, the platform detects entire fraud rings, identifies the ringleader through whom money is funnelled, traces the money trail hop by hop, maps how a single ring spans multiple districts and states, answers an investigator's natural-language questions through a tool-calling AI agent, and — with one click — seals a tamper-evident, court-admissible intelligence package.

The platform's central thesis is a shift in posture: from reactive case investigation, which begins only after a victim files a complaint and the money is gone, to proactive network detection that surfaces a ring while it is still active and quantifies exactly how much harm earlier intervention would have prevented. Every claim the system makes is explainable and evidence-backed; no output is a black-box score.

What was built and verified: a complete, running end-to-end system — a synthetic-data generator with planted, verifiable fraud rings; an in-process detection engine using real graph algorithms; a FastAPI backend with a hash-chained evidence ledger; a tool-calling LLM investigation agent; and a polished React command centre. Detection results were validated against known ground truth, not merely asserted.

2. Problem Statement & Context

India is facing an industrial-scale escalation in digital financial crime. The scale and the shape of the problem both matter:

1.14 million cybercrime complaints were registered in India in 2023 — an increase of **60%** year over year (Ministry of Home Affairs).

“Digital arrest” scams — where fraudsters impersonating CBI, ED, or Customs officers hold victims in multi-day psychological hostage situations over video call — defrauded citizens of over **₹1,776 crore** in just the first nine months of 2024 (Ministry of Home Affairs).

The RBI's Annual Report 2025 flagged record seizures of high-quality Fake Indian Currency Notes (FICN) of sufficient quality to defeat manual detection in routine banking.

Critically, these are not opportunistic crimes committed by lone actors. They are industrialised operations run from fraud compounds, using spoofed numbers, mule-account networks, and fake government portals, frequently coordinated across state and even national borders. A victim in one district, a mule account in a second, and a cash-out point in a third are all facets of a single operation — but each local police station only ever sees its own fragment.

The core gap: what investigators lack is not evidence after the fact — complaints eventually arrive. What they lack is intelligence before mass victimisation, and tooling that reconstructs the whole network from those fragments rather than examining one flagged transaction at a time. CyberInvestigator is built specifically to close that gap.

3. Solution Overview

The platform delivers seven core capabilities, each mapped to a concrete investigator need:

1. **Fraud-ring detection.** Graph community detection (Louvain) clusters accounts that share devices, UPI handles, and money flow into a single identified ring — surfacing the network, not a list of isolated flags.
2. **Ringleader identification.** Betweenness centrality on the directed money-flow graph pinpoints the entity through which the money funnels — the operational hub, not merely the busiest account.
3. **Explainable risk scoring.** A transparent rule engine, an unsupervised anomaly detector (IsolationForest), and a supervised model (RandomForest) are fused into one score, and every flag carries human-readable reasons.
4. **Lead-time reconstruction.** The system computes when a ring first became detectable versus how many victims were defrauded afterward — turning detection into a quantified prevention argument.

5. **Cross-jurisdiction mapping.** DBSCAN geo-clustering plus a live map and rotating-globe view reveal a ring's footprint across districts and states.
6. **Live investigation agent.** A tool-calling LLM agent answers questions by querying the case graph in real time and citing entity and transaction IDs for every claim; it degrades to a deterministic answer when no API key is present.
7. **Court-admissible evidence.** One click seals a SHA-256 hash-chained Intelligence Package that is tamper-evident and auditable.

4. System Architecture

CyberInvestigator is a modular monolith with clean internal boundaries, designed so that no single component is a hard dependency of the core detection loop. Data flows linearly and auditably:

Synthetic Dataset (persons · accounts · UPI · phones · devices · IPs · transactions)

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Validation → **Entity Resolution** → **Feature Engineering**

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Scoring Engine (Rule Engine + IsolationForest + RandomForest → Score Fusion) | **Graph Engine** (NetworkX: Louvain · Betweenness · WCC · shortest-path · DBSCAN)

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FastAPI Investigation API + SHA-256 Hash-Chained Ledger → (optional) Neo4j Aura for graph storage / visualisation

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React Command Centre (graph · map · globe · evidence) and **Tool-Calling Agent** (OpenRouter)

Design decision 1 — In-process detection, no mandatory database

Detection runs entirely in-process using NetworkX, pandas, and scikit-learn. Neo4j Aura's free and standard tiers do not ship the Graph Data Science (GDS) library, so rather than take a hard dependency on a paid tier, every graph algorithm runs directly against the data in Python. Neo4j is wired in only as an optional storage and visualisation layer. The practical consequence: the platform boots and detects fraud rings correctly with zero external services — a deliberate reliability and portability choice, not a limitation.

Design decision 2 — The LLM is a tool-user, not a summariser

The agent is never handed a raw data dump to summarise. It is given a fixed set of read-only tools and must decide which to call and cite specific IDs for every claim. Classification and scoring remain deterministic and reproducible in the engine; the LLM only reasons over already-computed evidence. This keeps the system auditable and prevents the model from inventing findings.

5. Dataset Design

Because the platform is demonstrated on synthetic data, the dataset was engineered so that detection has genuine, verifiable signal to find — not a trivially separable toy set. A deterministic, seed-based generator produces roughly 1,500 persons and 15,000 transactions across the following interlinked entity types, embedded in realistic, India-localised background activity:

Entity	Role in the graph
Person	Individual actor — victim, mule, scammer, ringleader, or ordinary user
Bank Account	Owned by a person; source/target of money transfers
UPI Handle	Payment identifier; a handle reused across accounts is a mule signal

Entity	Role in the graph
Phone	May be flagged as spoofed (a scammer signal)
Device	A single device used by many accounts indicates a mule farm
IP	Login origin; shared IPs cluster related actors
Transaction	Directed money transfer between accounts, with amount and timestamp

Into this background the generator plants three complete fraud rings with deliberately discoverable structure: a designated ringleader through whom money funnels, mule accounts sharing devices and UPI handles, a multi-hop money trail from victim to cash-out, victims spread across multiple districts and states with mules concentrated in one or two, transfers structured just under the ₹50,000 reporting threshold, and precise timestamps that encode when each ring first became detectable.

Verifiability by construction. The generator runs an 8-point acceptance report on every execution — confirming, for example, that the planted ringleader genuinely has the highest betweenness centrality, that a device is shared by at least eight mule accounts, that a valid victim→ringleader→cash-out path exists, and that the geographic spread matches the cross-jurisdiction story. If any check fails, the data is rejected. A companion GROUND_TRUTH file records the exact answer key, so every detection result can be measured against known truth rather than eyeballed.

6. Detection Methodology

6.1 Entity resolution & feature engineering

Entities are resolved deterministically (normalised identifiers) rather than via fuzzy matching, keeping the pipeline reproducible. Each person is then described by a vector of behavioural and graph-derived features, including:

Feature	What it captures
Transaction velocity (in/out counts)	Bursty, machine-like activity versus normal cadence
Fan-in / fan-out ratio	The mule signature — receives from many, forwards to few
Graph degree & connectivity	How embedded an entity is in the network
Off-hours activity ratio	Concentration of activity in the 00:00–05:00 window
Structuring count	Transfers deliberately just under ₹50,000
Account age at activity	Freshly opened accounts already transacting heavily
Shared-device / reused-UPI flags	Direct infrastructure links to other flagged accounts
Spoofed-phone flag	Association with a spoofed number

6.2 Scoring engine (explainable, three-signal fusion)

Three independent signals are combined so that no single method is a point of failure and every score remains explainable:

- **Rule engine** — a transparent, additive set of expert rules, each contributing points and a human-readable reason (e.g. “Shares a device with multiple other accounts”, “High fan-in / low fan-out — mule pattern”, “Transfers just under ₹50,000”, “Linked to a spoofed phone number”).
- **Anomaly detection** — an IsolationForest surfaces entities whose overall behaviour is statistically unusual, catching novel patterns the rules were not written for.
- **Supervised model** — a RandomForest trained on the labelled data provides a learned probability of being a perpetrator.

The three are fused into a single 0–1 score with the deliberate constraint that the fused score never falls below what the transparent rules alone would assign — so explainability is never sacrificed to the ML component. Every entity is banded Low / Medium / High and carries its reason list.

6.3 Graph algorithms

Task	Algorithm & note
Detect the fraud ring	Louvain community detection over an entity graph combining money edges, shared-device edges, and shared-UPI edges
Identify the ringleader	Betweenness centrality on the directed money-flow graph (the funnel), not high in-degree
Isolate mule networks	Weakly connected components
Trace the money trail	Shortest path over money edges: victim → mule → mule → ringleader → cash-out
Find geographic hotspots	DBSCAN on latitude/longitude with a haversine metric, independent of admin boundaries

A note on rigour: two candidate approaches were tested and rejected on evidence. Ranking entities by PageRank surfaces high-in-degree mules rather than the ringleader, so betweenness on the directed money graph is used instead; and feeding raw graph degree into the supervised classifier did not improve per-entity detection, so the graph's value is presented honestly as ring-level intelligence rather than as a booster for the transaction score.

7. The Investigation Agent

An investigator can simply ask questions in natural language. The agent is implemented over OpenRouter (making it model-agnostic — the underlying model is a single configuration value) and is constrained to answer only by calling a fixed set of read-only tools:

- **get_entity** — risk scores, reasons, and features for a given entity
- **get_ring** — members, ringleader, and statistics for a detected ring
- **get_shared_devices** — the shared-device evidence linking an entity to others
- **money_trail** — the shortest money path between two entities
- **list_rings** — all detected rings ranked by risk

Every answer must cite the specific entity and transaction IDs it relied on, and the interface shows which tools were called. If no API key is configured, the endpoint returns a deterministic, still evidence-grounded answer rather than failing — so the platform is fully demonstrable with zero external cost.

8. Court-Admissible Evidence

The brief explicitly asks that intelligence packages be auditable for legal admissibility. CyberInvestigator answers this directly. Each sealed Intelligence Package contains the ring summary, the lead-time statement, the ringleader together with the evidentiary basis for that designation, every flagged entity with its reasons, the traced money path, and the list of algorithms used to produce the finding.

The package is serialised deterministically and hashed with SHA-256, and — crucially — each new package's hash is chained to the hash of the previous one, forming a tamper-evident ledger. If any past package is altered, its stored hash no longer matches a recomputation, and every subsequent link in the chain breaks. A verification endpoint recomputes the entire chain on demand and reports the first broken link, if any. This provides a concrete, checkable guarantee that an evidence bundle has not been modified after it was generated.

9. Technology Stack

Layer	Choice	Rationale
Frontend	React 18, TypeScript, Vite, Tailwind v4	Fast iteration, type safety, no CSS build friction
Graph viz	Cytoscape.js	Renders directly from JSON; no graph DB required
Map / geo	Leaflet + OpenStreetMap	No API key, no vendor lock-in
Backend	FastAPI (Python)	Modular monolith, one process to run
Detection	NetworkX, pandas, scikit-learn	All required algorithms free; no GDS dependency
ML models	RandomForest + IsolationForest + rules	Explainable and small-data friendly
Graph DB (optional)	Neo4j Aura	Storage / Bloom visualisation only
LLM	OpenRouter (tool-calling)	Model-agnostic; swap models via one config value
Ledger	SHA-256 hash chain (flat JSON)	Tamper-evident audit trail, zero extra infra

10. Evaluation & Validated Results

Results were measured against the generator's known ground truth, not asserted from a single demo run:

Metric	Result
Ring-detection purity (Louvain)	Recovers the planted hero ring at 100% purity
Supervised model (PR-AUC)	0.77 — realistic, after a data leak was found and removed (see below)
Entities analysed	1,500
Fraud rings detected	3
Entities flagged high-risk	33
Total amount at risk identified	₹1.76 crore
Lead-time example	Ring detectable at victim #3; 62 further victim payments occurred afterward

Why PR-AUC, and why 0.77 rather than a suspicious 1.00. Fraud is a heavily imbalanced problem, so precision-recall AUC is the honest metric (plain accuracy would look inflated and mean little). During testing, the model initially scored a near-perfect PR-AUC — a red flag. Investigation revealed a data leak: account age alone almost perfectly separated the classes, letting the model take a shortcut. The dataset was corrected so that account age overlaps between classes, forcing the model to rely on genuine behavioural and network signal. The resulting 0.77 is a realistic, defensible figure — and the story itself demonstrates methodological rigour.

The lead-time result, in plain terms. For the demonstrated ring, the shared infrastructure that betrays the network became observable at the third victim — yet 62 further victim payments were made after that point. Acting at the moment of detectability rather than waiting for complaints would have prevented the large majority of the harm. This is the platform's central business-impact claim, expressed as a number.

11. Key Differentiators

- Explainable AI — a human-readable reason accompanies every flag, never a bare black-box score.

- A tool-calling investigation agent that reasons over live evidence and cites IDs, rather than summarising a static document.
- Tamper-evident, hash-chained intelligence packages purpose-built for legal admissibility.
- Cross-jurisdiction visualisation of rings that span multiple states — visibility no single police station has in isolation.
- A lead-time metric that quantifies prevented harm, not merely detection accuracy.
- Runs with zero external services — detection has no hard dependency on any database or paid API.

12. User Personas & Use Cases

- **Cybercrime investigator (LEA).** Uploads a batch of complaints, sees them auto-clustered into rings, identifies the ringleader, and exports a court-ready evidence package — collapsing days of manual link-analysis into minutes.
- **Bank / fintech fraud analyst.** Triage flagged accounts with explainable reasons and a risk band, and traces where funds moved before freezing or reporting them.
- **Inter-agency coordinator.** Uses the cross-jurisdiction map to see that a ring spans multiple states and route a consolidated intelligence package to the right jurisdictions simultaneously.

13. Security, Privacy & Ethics

- Synthetic data only — the demonstration uses no real personal, financial, or law-enforcement data.
- Secrets are held in environment configuration, never committed to source control.
- All database and graph queries are parameterised; user-supplied search terms are never string-concatenated into queries.
- The hash-chained ledger provides a tamper-evident chain of custody for generated intelligence — a security property, not just a feature.
- Human-in-the-loop by design — the platform surfaces evidence and recommendations; a human investigator makes every consequential decision.
- **For production,** authentication, role-based access control, PII minimisation and masking, data-retention governance, and formal legal review of the evidence-package format would be required before handling real citizen data.

14. Scope Decisions

The brief lists five illustrative tracks: Digital Arrest Scam Detection, Counterfeit Currency Identification, Fraud Network Graph Intelligence, Geospatial Crime Pattern Intelligence, and Citizen Fraud Shield. Rather than build shallow versions of all five, the team went deep on Fraud Network Graph Intelligence as the primary track, with Geospatial Crime Pattern Intelligence as a tightly-integrated secondary track that reuses the same data and detection results. This was a deliberate strategy to maximise technical depth, correctness, and judge-visible rigour within the build window — a focused, defensible product rather than a broad but thin one.

15. Judging Criteria Alignment

Criterion	Weight	How the project addresses it
Innovation	25%	Tool-calling, evidence-citing agent combined with hash-chained, court-admissible intelligence packages
Business Impact	25%	The lead-time metric — a quantified “victims prevented if acted earlier” number, not a vague claim
Technical Excellence	20%	Real graph algorithms (Louvain, betweenness, WCC, shortest-path, DBSCAN) and explainable fused scoring, verified against ground truth

Criterion	Weight	How the project addresses it
Scalability	15%	Modular monolith with a clear microservice-split path; batch pipeline today, queue-based ingestion as the documented production path
User Experience	15%	One coherent command centre combining graph, map, agent, and evidence in a single investigative flow

16. Challenges & Learnings

- **A hidden data leak.** A near-perfect early model score exposed an account-age leak; fixing the dataset to force reliance on real signal was a key methodological lesson.
- **Choosing the right graph algorithm.** Testing showed PageRank surfaces mules, not the ringleader — betweenness on the directed money graph is the correct tool, a distinction confirmed empirically rather than assumed.
- **Working within Neo4j Aura’s limits.** The absence of the GDS library on the free tier drove the decision to run detection in-process with NetworkX — which turned into a strength: zero mandatory external services.
- **Robust visuals over fragile ones.** An initial WebGL globe proved unstable across environments, so it was rebuilt on dependency-free Canvas2D — prioritising reliability that will hold up on any judge’s machine.

17. Roadmap

- Replace RandomForest with XGBoost and add SHAP for per-feature attribution charts in the entity view.
- Optionally read the graph view live from Neo4j when a GDS-enabled instance is available.
- Add a citizen-facing lightweight fraud-check widget reusing the same backend and scoring.
- Move to queue-based ingestion to support national-scale data volume.

18. How to Run (Reproducibility)

The entire system is reproducible from the repository with standard tooling. In brief: create a Python environment and install backend requirements; run the data generator (which prints its acceptance report); start the FastAPI backend; then install and start the React frontend, which proxies to the backend. The LLM agent and Neo4j are optional — without either, the platform still runs and detects rings. Full step-by-step instructions, the API reference, and a component-by-component map are documented in the repository README and AGENTS.md.

19. Disclaimer

Built for a national AI hackathon on entirely synthetic data. The platform is not connected to any real financial system, law-enforcement database, or citizen data, and is not intended for production use as-is. Statistics cited in the Problem Statement are drawn from the referenced public sources.

20. Repository & Resources

GitHub repository: github.com/AayushG10/cyber-intelligence-knowledge-base-investigator

The repository contains the full source code (backend, frontend, and data generator), an editable architecture diagram in Excalidraw format, a presentation deck, a shot-by-shot demo-video recording script, and both PDF and editable versions of this submission document. A detailed README and an AGENTS.md design-rationale document provide complete onboarding and the reasoning behind every technical decision.